Quivers

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Abstract

These notes are based on the lecture by Victor Ginzburg at the University of Chicago in summer 2026. We introduce quivers and their finite-dimensional representations, describe representation spaces and the base-change action, state the finite-type form of Gabriel's theorem, and explain the Tits-form argument showing that finite representation type forces the underlying graph to be a simply laced Dynkin diagram.

Contents

1 Quivers and Their Representations

Throughout, let k be a field.

Definition 1.1. A *quiver* is a directed graph

$$Q = (I, E, s, t),$$

where I is the set of vertices, E is the set of arrows, and

$$s, t : E \longrightarrow I$$

assign to each arrow its source and target. An arrow $a \in E$ with $s(a) = i$ and $t(a) = j$ is written

$$a : i \longrightarrow j.$$

Unless stated otherwise, all quivers in these notes are finite.

Definition 1.2. A finite-dimensional representation ρ of a quiver Q over k consists of

(i) a finite-dimensional k-vector space V_i of dimension $V_i = d_i$ for every vertex $i \in I$;

(ii) a k-linear map

$$\rho(a) : V_i \longrightarrow V_j$$

for every arrow $a : i \to j$.

The vector

$$d = \dim V = (d_i)_{i \in I} \in \mathbb{Z}^I_{\geqslant 0}$$

is called the *dimension vector* of V.

Definition 1.3. Let V and V' be representations of Q. A *morphism*

$$f : V \longrightarrow V'$$

is a collection of linear maps $f_i : V_i \to V'_i$ such that, for every arrow $a : i \to j$,

$$f_j \circ \rho(a) = \rho'(a) \circ f_i. \tag{1.1}$$

Equivalently, the following square commutes:

$$\begin{tikzcd} V_i \arrow[r, "\rho(a)"] \arrow[d, "f_i"'] & V_j \arrow[d, "f_j"] \\ V'_i \arrow[r, "\rho'(a)"'] & V'_j \end{tikzcd}$$

The morphism f is an *isomorphism* if and only if every f_i is invertible.

Example 1.4. Consider the quiver

$$\begin{tikzcd} 1 \arrow[loop above, "a"] \end{tikzcd}$$

A representation is exactly a linear map $A : V \to V$. If another representation is given by $B : V' \to V'$, then a morphism from A to B is a pair (f_1, f_2) satisfying

$$f_2 A = B f_1.$$

Thus two representations are isomorphic precisely when their corresponding matrices are related by independent changes of bases in the domain and codomain.

2 Dynkin Diagrams and Gabriel's Theorem

The connected simply laced Dynkin diagrams are of types A_n, D_n, E_6, E_7, and E_8.

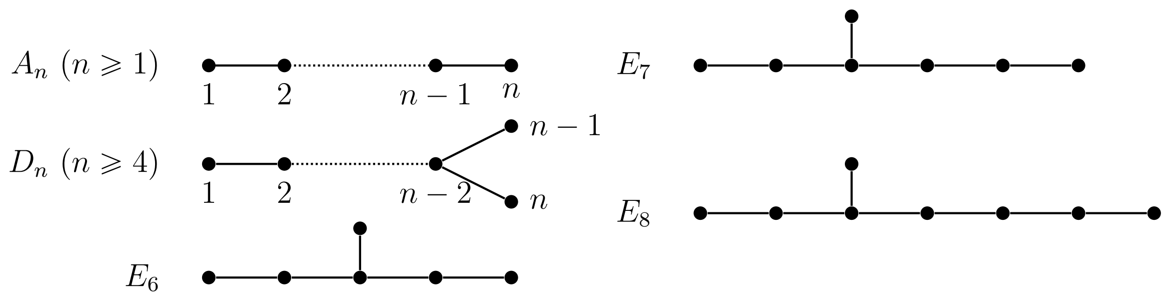

Figure 1: The connected simply laced Dynkin diagrams.

Theorem 2.1 (Gabriel)**.** *Let k be algebraically closed and let Q be a connected quiver. The following are equivalent:*

(1) *the underlying unoriented graph of Q is one of the Dynkin diagrams A_n, D_n, E_6, E_7, or E_8;*

(2) *for every dimension vector $\boldsymbol{d}$, there are only finitely many isomorphism classes of representations of Q of dimension vector $\boldsymbol{d}$.*

Remark 2.2. The same simply laced Dynkin diagrams also appear in the classification of finite-dimensional complex simple Lie algebras and, through the McKay correspondence, in the classification of finite subgroups of $\mathrm{SL}_2(\mathbb{C})$ up to conjugacy. This recurring phenomenon is often called the *ADE correspondence.*

3 Representation Spaces and Base Change

Fix a dimension vector $\boldsymbol{d} = (d_i)_{i \in Q_0}$ and choose standard vector spaces

$$V_i = k^{d_i}.$$

Definition 3.1. The representation space of Q with dimension vector $\boldsymbol{d}$ is the affine space

$$\mathrm{Rep}(Q, \boldsymbol{d}) = \prod_{a:i \to j} \mathrm{Hom}_k(V_i, V_j). \tag{3.1}$$

Consequently,

$$\dim \mathrm{Rep}(Q, \boldsymbol{d}) = \sum_{a:i \to j} d_i d_j. \tag{3.2}$$

Define the base-change group

$$G_{\boldsymbol{d}} = \prod_{i \in Q_0} \mathrm{GL}(V_i). \tag{3.3}$$

For $g = (g_i)_{i \in Q_0} \in G_{\boldsymbol{d}}$ and $V = (V_a)_{a \in Q_1} \in \mathrm{Rep}(Q, \boldsymbol{d})$, set

$$g\rho(a) = g_j \rho(a) g_i^{-1} \qquad \text{for every arrow } a : i \to j. \tag{3.4}$$

This is the usual change-of-basis action.

Proposition 3.2. *Two points of* $\mathrm{Rep}(Q, \boldsymbol{d})$ *lie in the same* $G_{\boldsymbol{d}}$*-orbit if and only if the corresponding representations are isomorphic. Hence the isomorphism classes of representations with dimension vector* $\boldsymbol{d}$ *are exactly the* $G_{\boldsymbol{d}}$*-orbits in* $\mathrm{Rep}(Q, \boldsymbol{d})$.

Proof. An element $g = (g_i)$ identifies two representations precisely when each g_i is an isomorphism and

$$g_j \rho(a) = \rho'(a) g_i$$

for every arrow $a : i \to j$. This is equivalent to $\rho'(a) = g_j \rho(a) g_i^{-1}$, which is exactly the orbit relation in (3.4). □

4 A One-Parameter Family for $\widetilde{D}_4$

The following example explains geometrically why non-Dynkin quivers may have infinitely many isomorphism classes in a fixed dimension vector.

Let Q be the quiver

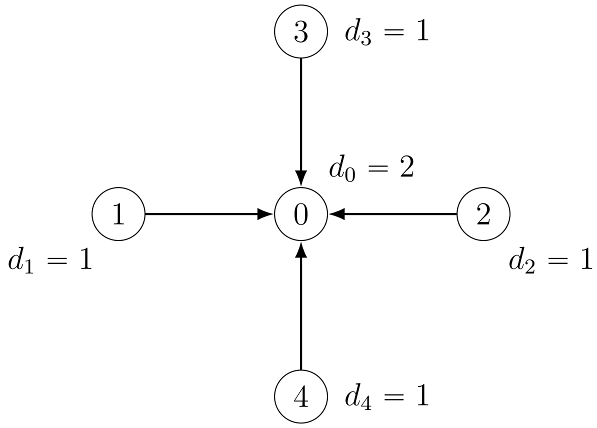

Figure 2: An orientation of the affine Dynkin graph $\widetilde{D}_4$.

Thus a representation of dimension vector $(2, 1, 1, 1, 1)$ is given by four maps

$$A_i : k \longrightarrow k^2, \qquad i = 1, 2, 3, 4.$$

Assume that every A_i is injective. Then

$$\ell_i = \mathrm{Im}(A_i)$$

is a line in k^2, so the representation determines an ordered quadruple

$$(\ell_1, \ell_2, \ell_3, \ell_4) \in (\mathbb{P}^1(k))^4.$$

The factors $\mathrm{GL}_1(k)$ at the four outer vertices merely rescale the maps A_i, while the factor $\mathrm{GL}_2(k)$ at the central vertex acts simultaneously on the four lines. Therefore, isomorphism classes are ordered quadruples of points of $\mathbb{P}^1(k)$ modulo the diagonal action of $\mathrm{PGL}_2(k)$.

For $\lambda \in k\backslash\{0,1\}$, define a representation $V(\lambda)$ by

$$\begin{aligned} A_1(1) &= (1,0), \quad A_2(1) = (0,1), \\ A_3(1) &= (1,1), \quad A_4(1) = (\lambda,1). \end{aligned} \tag{4.1}$$

The four image lines correspond to the ordered points

$$(\infty, 0, 1, \lambda) \in \mathbb{P}^1(k).$$

Their cross-ratio is

$$[\infty, 0; 1, \lambda] = \lambda. \tag{4.2}$$

Since the cross-ratio is invariant under $\mathrm{PGL}_2(k)$, an isomorphism

$$V(\lambda) \cong V(\mu)$$

forces $\lambda = \mu$. Because an algebraically closed field is infinite, this gives infinitely many pairwise nonisomorphic representations of the same dimension vector.

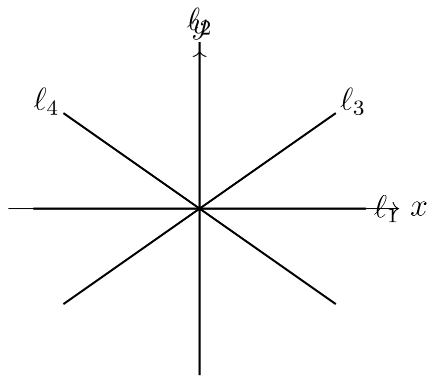

Figure 3: Four ordered lines through the origin in k^2; their projective configuration is measured by the cross-ratio.

5 The Tits Form and the Easy Direction of Gabriel's Theorem

We now prove the implication

$$\text{finite representation type} \Longrightarrow \text{Dynkin underlying graph.}$$

The converse is the deeper direction of Gabriel's theorem.

Assume that Q has no loops. For distinct vertices $i, j \in I$, let $h_{ij} = h_{ji}$ be the number of arrows whose endpoints are i and j, with orientation forgotten. Set $h_{ii} = 0$.

Definition 5.1. Let $\{e_i\}_{i\in I}$ be the standard basis of $\mathbb{R}^I$. Define a symmetric bilinear form by

$$(e_i, e_j)_Q = \begin{cases} 2, & i = j, \\ -h_{ij}, & i \neq j. \end{cases} \tag{5.1}$$

The associated Tits quadratic form is

$$q_Q(\boldsymbol{x}) = \frac{1}{2}(\boldsymbol{x}, \boldsymbol{x})_Q = \sum_{i\in Q_0} x_i^2 - \sum_{\{i,j\}\subseteq Q_0} h_{ij} x_i x_j. \tag{5.2}$$

Equivalently,

$$q_Q(\boldsymbol{d}) = \sum_i d_i^2 - \sum_{a:i\to j} d_i d_j.$$

Proposition 5.2. *Suppose that* $\mathrm{Rep}(Q, \boldsymbol{d})$ *contains only finitely many* $G_{\boldsymbol{d}}$*-orbits for a nonzero dimension vector* $\boldsymbol{d}$*. Then*

$$q_Q(\boldsymbol{d}) \geqslant 1. \tag{5.3}$$

Proof. The affine space $\mathrm{Rep}(Q, \boldsymbol{d})$ is irreducible. If it is a finite union of $G_{\boldsymbol{d}}$-orbits, one of those orbits is dense. Since algebraic-group orbits are locally closed, a dense orbit is open. Let $\mathcal{O}$ be such an open orbit and choose $V \in \mathcal{O}$. Then

$$\dim \mathcal{O} = \dim G_{\boldsymbol{d}} - \dim(G_{\boldsymbol{d}})_V, \tag{5.4}$$

where $(G_{\boldsymbol{d}})_V$ is the stabilizer of V.

Because $\mathcal{O}$ is open in $\mathrm{Rep}(Q, \boldsymbol{d})$,

$$\dim \mathcal{O} = \dim \mathrm{Rep}(Q, \boldsymbol{d}) = \sum_{a:i\to j} d_i d_j.$$

Moreover,

$$\dim G_{\boldsymbol{d}} = \sum_i d_i^2.$$

The one-dimensional subgroup of simultaneous scalar transformations

$$\{(\lambda I_{V_i})_{i\in Q_0} : \lambda \in k^\times\}$$

fixes every representation, so

$$\dim(G_{\boldsymbol{d}})_V \geqslant 1.$$

Combining these equalities and inequalities gives

$$\sum_{a:i\to j} d_i d_j \leqslant \sum_i d_i^2 - 1,$$

which is equivalent to $q_Q(\boldsymbol{d}) \geqslant 1$. $\square$

Lemma 5.3. *If* $q_Q(\boldsymbol{d}) > 0$ *for every nonzero vector* $\boldsymbol{d} \in \mathbb{Z}_{\geqslant 0}^{Q_0}$*, then* q_Q *is positive definite on* $\mathbb{R}^{Q_0}$*.*

Proof. Suppose that q_Q is not positive definite. If there exists $\boldsymbol{x} \neq 0$ with $q_Q(\boldsymbol{x}) < 0$, then, because all off-diagonal coefficients of q_Q are nonpositive,

$$q_Q(|\boldsymbol{x}|) \leqslant q_Q(\boldsymbol{x}) < 0,$$

where $|\boldsymbol{x}| = (|x_i|)_i$. By continuity, a sufficiently close nonzero rational vector $\boldsymbol{y} \in \mathbb{Q}_{\geqslant 0}^{Q_0}$ also satisfies $q_Q(\boldsymbol{y}) < 0$. Clearing denominators yields a nonzero $\boldsymbol{d} \in \mathbb{Z}_{\geqslant 0}^{Q_0}$ with $q_Q(\boldsymbol{d}) < 0$, a contradiction.

It remains to consider the case in which q_Q is positive semidefinite but singular. The matrix of $(\cdot,\cdot)_Q$ has integer entries and nontrivial kernel, so its kernel contains a nonzero rational vector $\boldsymbol{z}$. Then $q_Q(\boldsymbol{z}) = 0$. Again,

$$q_Q(|\boldsymbol{z}|) \leqslant q_Q(\boldsymbol{z}) = 0.$$

Positive semidefiniteness forces $q_Q(|\boldsymbol{z}|) = 0$. After clearing denominators, we obtain a nonzero vector $\boldsymbol{d} \in \mathbb{Z}_{\geqslant 0}^{Q_0}$ satisfying $q_Q(\boldsymbol{d}) = 0$, again a contradiction. Therefore q_Q is positive definite. □

Proposition 5.4. *Let* Γ *be a finite connected graph, allowing multiple edges but no loops. Its symmetric Tits form is positive definite if and only if* Γ *is one of the simply laced Dynkin diagrams*

$$A_n, \qquad D_n, \qquad E_6, \qquad E_7, \qquad E_8.$$

In particular, positive definiteness excludes multiple edges.

Remark 5.5. Indeed, if two vertices are joined by $m \geqslant 2$ edges, then the corresponding 2×2 principal submatrix is

$$\begin{pmatrix} 2 & -m \\ -m & 2 \end{pmatrix},$$

whose determinant is $4 - m^2 \leqslant 0$. Hence a positive-definite symmetric Tits form can only arise from a simple graph.

Proof of Theorem 2.1, (2)⇒(1). Assume that, for every dimension vector $\boldsymbol{d}$, the variety $\operatorname{Rep}(Q, \boldsymbol{d})$ has only finitely many isomorphism classes. By Proposition 3.2, it has only finitely many $G_{\boldsymbol{d}}$-orbits. Proposition 5.2 therefore gives

$$q_Q(\boldsymbol{d}) \geqslant 1 \qquad \text{for every nonzero } \boldsymbol{d} \in \mathbb{Z}_{\geqslant 0}^{Q_0}.$$

By Lemma 5.3, the Tits form q_Q is positive definite. Proposition 5.4 then implies that the underlying unoriented graph of Q is a simply laced Dynkin diagram. □